

Multiple Choice. Select the letter of your answer

1. Which inequality gives the domain of $y = \sqrt{2x - 3}$?

A. $x < \frac{3}{2}$

C. $x \leq \frac{3}{2}$

B. $x > \frac{3}{2}$

D. $x \geq \frac{3}{2}$

$$2x - 3 \geq 0$$

$$\mathbf{D: } x \geq \frac{3}{2}$$

2. The function $f(x) = x^3$ is transformed to $g(x) = (x - 2)^3 + 5$ by a horizontal translation of 2 units followed by a vertical translation of 5 units.

Which row of the table shows the directions of the translations?

	Horizontal translation of 2 units	Vertical translation of 5 units
A.	Left	Up
B.	Right	Up
C.	Left	Down
D.	Right	Down

B: Right, Up

3. The probability distribution table for a discrete random variable X is shown.

What is the expected value of X ?

A. 0.6

B. 1.0

C. 1.5

D. 2.0

$$E(X) = 1(0.6) + 2(0.3) + 3(0.1) = 1.5$$

C: 1.5

x	$P(X = x)$
1	0.6
2	0.3
3	0.1

4. Which of the following is the range of the function $f(x) = x^2 - 1$?

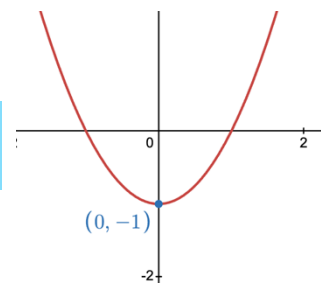
A. $[-1, \infty)$

C. $[-1, 1]$

B. $(-\infty, 1]$

D. $(-\infty, \infty)$

A: $[-1, \infty)$



5. What is $\int \frac{1}{(2x + 1)^2} dx$?

A. $\frac{-2}{2x + 1} + C$

C. $2 \ln(2x + 1) + C$

B. $\frac{-1}{2(2x + 1)} + C$

D. $\frac{1}{2} \ln(2x + 1) + C$

$$\int f'(x)[f(x)]^n dx = \frac{1}{n+1}[f(x)]^{n+1} + c$$

where $n \neq -1$

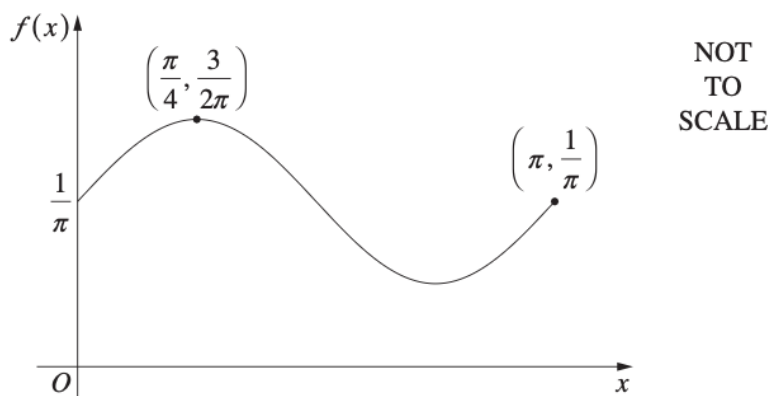
$$\mathbf{B: } -\frac{1}{2(2x+1)} + C$$

6. Consider the following graph of a probability density function $f(x)$.

What is the value of the mode?

- A. $\frac{1}{\pi}$
 B. $\frac{3}{2\pi}$
 C. $\frac{\pi}{4}$
 D. π

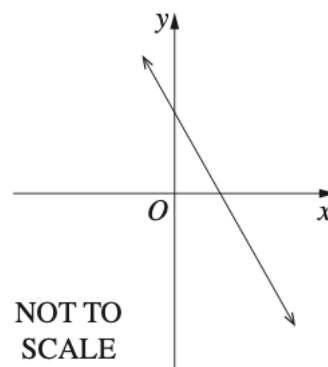
C: $\frac{\pi}{4}$



7. Which of the following could be the equation of this function?

- A. $y = 2x + 3$
 B. $y = 2x - 3$
 C. $y = -2x + 3$
 D. $y = -2x - 3$

C: $y = -2x + 3$



8. The parabola $y = (x - 3)^2 - 2$ is reflected about the y -axis. This is then reflected about the x -axis.

What is the equation of the resulting parabola?

- A. $y = (x + 3)^2 + 2$
 B. $y = (x - 3)^2 + 2$
 C. $y = -(x + 3)^2 + 2$
 D. $y = -(x - 3)^2 + 2$

C: $y = -(x + 3)^2 + 2$

9. What is $\int (6x + 1)^3 dx$?

- A. $\frac{1}{24}(6x + 1)^4 + C$
 B. $\frac{1}{4}(6x + 1)^4 + C$
 C. $\frac{2}{3}(6x + 1)^4 + C$
 D. $\frac{3}{2}(6x + 1)^4 + C$

A: $\frac{1}{24}(6x + 1)^4 + C$

10. Which interval gives the range of the function $y = 5 + 2\cos 3x$?

- A. $[2, 8]$
 B. $[3, 7]$
 C. $[4, 6]$
 D. $[5, 9]$

B: $[3, 7]$

Short Response Questions-Show your complete solutions.

11. Calculate the sum of the arithmetic series $4 + 10 + 16 + \cdots + 1354$.

$$T_n = a + (n - 1)d$$

$$1354 = 4 + (n - 1) \times 6$$

$$1350 = 6(n - 1)$$

$$n - 1 = 225 \implies n = 226$$

$$S_n = \frac{n}{2}(a + l) = \frac{226}{2}(4 + 1354) = 113 \times 1358 = 153454$$

12. Evaluate $\int_0^{\frac{\pi}{4}} \sec^2 x \, dx$.

$$\begin{aligned} \int_0^{\frac{\pi}{4}} \sec^2 x \, dx &= [\tan x]_0^{\frac{\pi}{4}} \\ &= \tan\left(\frac{\pi}{4}\right) - \tan(0) \\ &= 1 - 0 = 1 \end{aligned}$$

$$\int f'(x) \sec^2 f(x) \, dx = \tan f(x) + c$$

13. Find $\int \frac{x}{4 + x^2} \, dx$.

$$\int \frac{f'(x)}{f(x)} \, dx = \ln |f(x)| + c$$

$$= \frac{1}{2} \ln(4 + x^2) + C$$

14. (a) Differentiate $e^{2x}(2x + 1)$.

- Let $u = e^{2x}$, so $u' = 2e^{2x}$

- Let $v = 2x + 1$, so $v' = 2$

$$\frac{d}{dx} [e^{2x}(2x + 1)] = 2e^{2x}(2x + 1) + e^{2x}(2)$$

$$= e^{2x}(4x + 2 + 2) = e^{2x}(4x + 4) = 4(x + 1)e^{2x}$$

- (b) Hence, find $\int (x + 1)e^{2x} \, dx$.

$$\int (x + 1)e^{2x} \, dx = \frac{1}{4}e^{2x}(2x + 1) + C$$

15. Prove that $\sec \theta - \cos \theta = \sin \theta \tan \theta$.

$$\begin{aligned}
 \text{LHS} &= \sec \theta - \cos \theta = \frac{1}{\cos \theta} - \cos \theta \\
 &= \frac{1 - \cos^2 \theta}{\cos \theta} \\
 &= \frac{\sin^2 \theta}{\cos \theta} && \text{Since } 1 - \cos^2 \theta = \sin^2 \theta: \\
 &= \sin \theta \times \frac{\sin \theta}{\cos \theta} \\
 &= \sin \theta \tan \theta = \text{RHS}
 \end{aligned}$$

16. Evaluate $\int_{-2}^0 \sqrt{2x+4} \, dx$.

$$\int f'(x)[f(x)]^n \, dx = \frac{1}{n+1}[f(x)]^{n+1} + c$$

where $n \neq -1$

$$\begin{aligned}
 \int (2x+4)^{1/2} \, dx &= \frac{1}{2} \int 2(2x+4)^{1/2} \, dx \\
 &= \frac{1}{2} \left[\frac{1}{\frac{1}{2}+1} (2x+4)^{\frac{1}{2}+1} \right] + C = \frac{1}{2} \cdot \frac{2}{3} (2x+4)^{3/2} + C = \frac{1}{3} (2x+4)^{3/2} + C
 \end{aligned}$$

$$\int_{-2}^0 \sqrt{2x+4} \, dx = \left[\frac{1}{3} (2x+4)^{3/2} \right]_{-2}^0 = \frac{1}{3} (4^{3/2}) - \frac{1}{3} (0^{3/2}) = \frac{1}{3} (8) = \frac{8}{3}$$

17. The first term of an arithmetic sequence is 5. The sum of the first 43 terms is 2021.

What is the common difference of the sequence?

$$2021 = \frac{43}{2} [2(5) + 42d]$$

$$2021 = \frac{43}{2} (10 + 42d)$$

$$2021 = 43(5 + 21d)$$

$$\frac{2021}{43} = 5 + 21d$$

$$47 = 5 + 21d$$

$$42 = 21d \implies d = 2$$

18. Find the exact gradient of the tangent to the curve $y = x \tan x$ at the point where $x = \frac{\pi}{3}$.

$$\frac{dy}{dx} = (1) \tan x + x(\sec^2 x) = \tan x + x \sec^2 x$$

$$\tan\left(\frac{\pi}{3}\right) = \sqrt{3}, \quad \sec^2\left(\frac{\pi}{3}\right) = \left(\frac{1}{\cos(\pi/3)}\right)^2 = 4$$

$$x = \frac{\pi}{3}$$

$$\frac{dy}{dx} = \sqrt{3} + \frac{\pi}{3}(4) = \sqrt{3} + \frac{4\pi}{3}$$

19. The diagram shows a triangle ABC where $AC = 25$ cm, $BC = 16$ cm, $\angle BAC = 28^\circ$ and angle ABC is obtuse.

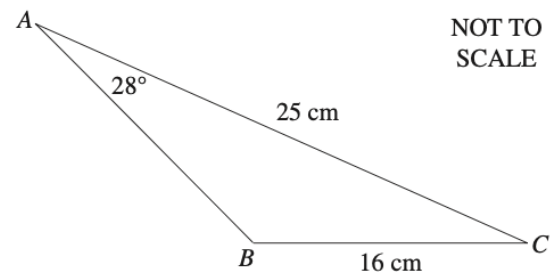
Find the size of the obtuse angle ABC correct to the nearest degree.

$$\frac{\sin B}{25} = \frac{\sin 28^\circ}{16}$$

$$\sin B = \frac{25 \times \sin 28^\circ}{16} \approx \frac{25 \times 0.4695}{16} \approx 0.7336$$

$$B \approx 47^\circ$$

$$\angle ABC = 180^\circ - 47^\circ = 133^\circ$$



20. A population, P , which is initially 5000, varies according to the formula $P = 5000b^{\frac{-t}{10}}$, where b is a positive constant and t is time in years, $t \geq 0$.

The population is 1250 after 20 years.

Find the value of t , correct to one decimal place, for which the instantaneous rate of decrease is 30 people per year.

When $t = 20$, $P = 1250$:

$$1250 = 5000b^{-2} \implies b^{-2} = 0.25$$

$$b^{-2} = 0.25 \implies b = 2$$

$$P = 5000(2)^{-t/10}.$$

$$\frac{dP}{dt} = 5000 \times \left(-\frac{1}{10} \ln 2\right) 2^{-t/10}$$

$$\frac{dP}{dt} = -30:$$

$$-500 \ln 2 \times 2^{-t/10} = -30$$

$$2^{-t/10} = \frac{30}{500 \ln 2} \approx 0.08656$$

$$-\frac{t}{10} \ln 2 = \ln(0.08656)$$

$$t = -10 \times \frac{\ln(0.08656)}{\ln 2} \approx 35.3$$

21. Initially there are 350 litres of water in a tank. Water starts flowing into the tank.

The rate of increase of the volume V of water in litres is given by $\frac{dV}{dt} = 300 - 7.5t$, where t is the time in hours.

Find the volume of water in the tank when $\frac{dV}{dt} = 0$.

1. **Find t when $\frac{dV}{dt} = 0$:**

$$300 - 7.5t = 0 \implies t = 40 \text{ hours}$$

2. **Integrate to find $V(t)$:**

$$V(t) = \int (300 - 7.5t) dt = 300t - 3.75t^2 + C$$

Initial volume $V(0) = 350$, so $C = 350$:

$$V(t) = 300t - 3.75t^2 + 350$$

3. **Evaluate at $t = 40$:**

$$V(40) = 300(40) - 3.75(40)^2 + 350 = 12000 - 6000 + 350 = 6350$$

22. What is the limiting sum of the following geometric series?

$$2000 - 1200 + 720 - 432 \dots$$

$$a = 2000, \quad r = \frac{-1200}{2000} = -0.6.$$

$$S_{\infty} = \frac{a}{1-r} = \frac{2000}{1-(-0.6)} = \frac{2000}{1.6} = 1250$$

23. A table of future value interest factors for an annuity of \$1 is shown.

<i>Rate</i> <i>Period</i>	1.5%	3%	4.5%	6%
5	5.152	5.309	5.471	5.637
10	10.703	11.464	12.288	13.181
20	23.124	26.870	31.371	36.786
40	54.268	75.401	107.030	154.762

- (a) Micky wants to save \$450 000 over the next 10 years.

If the interest rate is 6% per annum compounding annually, how much should Micky contribute each year? Give your answer to the nearest dollar.

annual contribution is R

$$450000 = R(13.181)$$

$$R = \frac{450000}{13.181}$$

$$\text{Contribution} = \frac{450000}{13.181} \approx 34145$$

- (b) Instead, Micky decides to contribute \$8535 every three months for 10 years to an annuity paying 6% per annum, compounding quarterly.

How much will Micky have at the end of 10 years?

$$\text{rate } i = \frac{0.06}{4} = 0.015 = 1.5\%.$$

$$\text{Number of quarters } n = 40.$$

$$\text{for } n = 40 \text{ at } 1.5\% \text{ is } 54.268.$$

$$FV = 8535(54.268) \approx 463177.38.$$

END

EXTRA QUESTION:

24. Consider the curve $y = ax^2 + bx + c$, where $a \neq 0$.

At a particular point, the tangent and normal to the curve are given by $t(x) = 2x + 3$ and $n(x) = -\frac{1}{2}x - 2$ respectively.

The curve has a minimum turning point at $x = -4$.

Find the values of a , b and c .

$$t(x) = 2x + 3, \quad n(x) = -\frac{1}{2}x - 2.$$

$$2x + 3 = -\frac{1}{2}x - 2$$

$$2x + \frac{1}{2}x = -5 \Rightarrow \frac{5}{2}x = -5 \Rightarrow x = -2$$

$$y = 2(-2) + 3 = -1$$

$$\boxed{(-2, -1)}$$

For $y = ax^2 + bx + c$,

$$\frac{dy}{dx} = 2ax + b.$$

$x = -2$, the tangent gradient is 2,

$$2a(-2) + b = 2 \Rightarrow -4a + b = 2 \quad (1)$$

Given the minimum turning point occurs at $x = -4$:

$$-\frac{b}{2a} = -4 \Rightarrow b = 8a \quad (2)$$

Substitute (2) into (1):

$$-4a + 8a = 2 \Rightarrow 4a = 2 \Rightarrow a = \frac{1}{2}.$$

$$b = 8a = 8 \cdot \frac{1}{2} = 4.$$

Find c using the point $(-2, -1)$

$$-1 = a(-2)^2 + b(-2) + c = 4a - 2b + c$$

$$-1 = 4\left(\frac{1}{2}\right) - 2(4) + c = 2 - 8 + c \Rightarrow c = 5.$$

$$\boxed{a = \frac{1}{2}, \quad b = 4, \quad c = 5.}$$

ANSWERS:

Question 15 (a)

Criteria	Marks
• Provides correct solution	2
• Identifies the correct factor from the table	1

Sample answer:

$$\begin{aligned}\text{Amount} &= \frac{\$450\,000}{13.181} \\ &= \$34\,140 \quad (\text{to the nearest dollar})\end{aligned}$$

Question 15 (b)

Criteria	Marks
• Provides correct solution	3
• Provides the correct interest rate and the correct number of periods, or equivalent merit	2
• Multiplies a factor from the table by \$8535, or equivalent merit	1

Sample answer:

$$\begin{aligned}r &= \frac{6}{4}\% \\ &= 1.5\%\end{aligned}$$

$$\begin{aligned}n &= 10 \times 4 \\ &= 40\end{aligned}$$

$$\begin{aligned}\text{Amount} &= \$8535 \times 54.268 \\ &= \$463\,177.38\end{aligned}$$

1. D
2. B
3. C
4. A
5. B
6. C
7. C
8. C
9. A
10. B

11. 153454
12. 1
13. $\frac{1}{2} \ln(4 + x^2) + C$
- 14(a). $4(x + 1)e^{2x}$
- 14(b). $\frac{1}{4}e^{2x}(2x + 1) + C$
14. $\sec \theta - \cos \theta = \sin \theta \tan \theta$
15. $\frac{8}{3}$
16. $d = 2$
17. $\sqrt{3} + \frac{4\pi}{3}$

19. $\angle ABC \approx 133^\circ$

20. $t \approx 35.3$ years

21. $V = 6350$ L

22. 1250

23(a). **\$34,140** per year

23(b). **\$463 177.38**

Working (brief teacher solutions)

1. $y = \sqrt{2x - 3} \Rightarrow 2x - 3 \geq 0 \Rightarrow x \geq \frac{3}{2} \rightarrow \mathbf{D}$
2. $g(x) = (x - 2)^3 + 5$: right 2, up 5 $\rightarrow \mathbf{B}$
3. $E(X) = 1(0.6) + 2(0.3) + 3(0.1) = 1.5 \rightarrow \mathbf{C}$
4. $f(x) = x^2 - 1$, minimum $-1 \rightarrow$ range $[-1, \infty) \rightarrow \mathbf{A}$
5. $\int (2x + 1)^{-2} dx$. Let $u = 2x + 1$: $\frac{1}{2} \int u^{-2} du = -\frac{1}{2u} + C = -\frac{1}{2(2x+1)} + C \rightarrow \mathbf{B}$
6. Mode = x -**value** where pdf is largest. Peak at $(\frac{\pi}{4}, \frac{3}{2\pi}) \rightarrow$ mode = $\frac{\pi}{4} \rightarrow \mathbf{C}$
7. Line has negative slope and positive y -intercept $\rightarrow y = -2x + 3 \rightarrow \mathbf{C}$
8. Reflect in y -axis: $y = (x + 3)^2 - 2$. Then in x -axis: $y = -(x + 3)^2 + 2 \rightarrow \mathbf{C}$
9. Let $u = 6x + 1$: $\int (6x + 1)^3 dx = \frac{1}{6} \int u^3 du = \frac{u^4}{24} + C \rightarrow \mathbf{A}$
10. $y = 5 + 2 \cos 3x \Rightarrow [5 - 2, 5 + 2] = [3, 7] \rightarrow \mathbf{B}$

Question	Your Answer	Correctness	Key Reason
1	D		Domain of $y = \sqrt{2x - 3}$ requires $2x - 3 \geq 0 \implies x \geq \frac{3}{2}$
2	B		$g(x) = (x - 2)^3 + 5$ is a right 2, up 5 translation of $f(x) = x^3$
3	C		$E(X) = 1(0.6) + 2(0.3) + 3(0.1) = 1.5$
4	A		$f(x) = x^2 - 1$ has minimum -1 , so range $[-1, \infty)$

5	B		$\int \frac{1}{(2x+1)^2} dx = -\frac{1}{2(2x+1)} + C$
6	C		Mode is the x -value at the peak of the PDF, which is $\frac{\pi}{4}$
7	C		The line has negative slope and positive y -intercept, matching $y = -2x + 3$
8	C		Reflections of $y = (x - 3)^2 - 2$ give $y = -(x + 3)^2 + 2$
9	A		$\int (6x + 1)^3 dx = \frac{1}{24}(6x + 1)^4 + C$
10	B		$y = 5 + 2 \cos 3x$ has range $[5 - 2, 5 + 2] = [3, 7]$

11. Sum of the arithmetic series $4 + 10 + 16 + \cdots + 1354$

Arithmetic series with $a = 4$, $d = 6$, $l = 1354$.

Find number of terms:

$$l = a + (n - 1)d$$

$$1354 = 4 + (n - 1)6 \Rightarrow 1350 = 6(n - 1) \Rightarrow n - 1 = 225 \Rightarrow n = 226$$

Sum:

$$S_n = \frac{n}{2}(a + l) = \frac{226}{2}(4 + 1354) = 113(1358) = 153454$$

153454

12. Evaluate $\int_0^{\pi/4} \sec^2 x \, dx$

$$\int \sec^2 x \, dx = \tan x + C$$

$$\int_0^{\pi/4} \sec^2 x \, dx = [\tan x]_0^{\pi/4} = 1 - 0 = 1$$

1

13. Find $\int \frac{x}{4 + x^2} dx$

Let $u = 4 + x^2 \Rightarrow du = 2x dx \Rightarrow x dx = \frac{1}{2} du$.

$$\int \frac{x}{4 + x^2} dx = \frac{1}{2} \int \frac{1}{u} du = \frac{1}{2} \ln |u| + C$$

$$\boxed{\frac{1}{2} \ln(4 + x^2) + C}$$

14.

(a) Differentiate $e^{2x}(2x + 1)$

Product rule:

$$\begin{aligned} \frac{d}{dx} (e^{2x}(2x + 1)) &= e^{2x} \cdot 2 + (2x + 1) \cdot 2e^{2x} \\ &= 2e^{2x}(1 + 2x + 1) = 4(x + 1)e^{2x} \end{aligned}$$

$$\boxed{4(x + 1)e^{2x}}$$

(b) Hence find $\int (x + 1)e^{2x} dx$

From (a), $\frac{d}{dx} (e^{2x}(2x + 1)) = 4(x + 1)e^{2x}$.

$$\int (x + 1)e^{2x} dx = \frac{1}{4} e^{2x}(2x + 1) + C$$

$$\boxed{\frac{1}{4} e^{2x}(2x + 1) + C}$$

15. Prove that $\sec \theta - \cos \theta = \sin \theta \tan \theta$

$$\sec \theta - \cos \theta = \frac{1}{\cos \theta} - \cos \theta = \frac{1 - \cos^2 \theta}{\cos \theta} = \frac{\sin^2 \theta}{\cos \theta} = \sin \theta \cdot \frac{\sin \theta}{\cos \theta} = \sin \theta \tan \theta$$

Proved

16. Evaluate $\int_{-2}^0 \sqrt{2x+4} dx$

Let $u = 2x + 4 \Rightarrow du = 2dx$, $dx = \frac{1}{2}du$.

When $x = -2$, $u = 0$; when $x = 0$, $u = 4$.

$$\int_{-2}^0 \sqrt{2x+4} dx = \frac{1}{2} \int_0^4 u^{1/2} du = \frac{1}{2} \left[\frac{2}{3} u^{3/2} \right]_0^4 = \frac{1}{2} \cdot \frac{2}{3} \cdot 8 = \frac{8}{3}$$

$\frac{8}{3}$

17. Arithmetic sequence: $a = 5$, $S_{43} = 2021$. Find d .

$$S_n = \frac{n}{2} (2a + (n-1)d)$$

$$2021 = \frac{43}{2} (10 + 42d)$$

$$4042 = 43(10 + 42d) = 430 + 1806d$$

$$3612 = 1806d \Rightarrow d = 2$$

$d = 2$

18. Gradient of tangent to $y = x \tan x$ at $x = \frac{\pi}{3}$

Product rule:

$$\frac{dy}{dx} = \tan x + x \sec^2 x$$

$$\text{At } x = \frac{\pi}{3}: \tan \frac{\pi}{3} = \sqrt{3}, \sec^2 \frac{\pi}{3} = \left(\frac{1}{\cos \frac{\pi}{3}} \right)^2 = 2^2 = 4.$$

$$m = \sqrt{3} + \frac{\pi}{3} \cdot 4 = \sqrt{3} + \frac{4\pi}{3}$$

$$\boxed{\sqrt{3} + \frac{4\pi}{3}}$$

Q19 (Trigonometry – sine rule)

Given $AC = 25$ cm, $BC = 16$ cm, $\angle BAC = 28^\circ$ and $\angle ABC$ is obtuse.

Let $\angle A = 28^\circ$. Side opposite $\angle A$ is $BC = 16$.

Side opposite $\angle B$ is $AC = 25$.

By the sine rule,

$$\frac{\sin B}{25} = \frac{\sin 28^\circ}{16} \Rightarrow \sin B = \frac{25 \sin 28^\circ}{16}.$$

$$\sin B \approx \frac{25(0.4695)}{16} \approx 0.7335.$$

So

$$B \approx \sin^{-1}(0.7335) = 47.2^\circ \quad \text{or} \quad 180^\circ - 47.2^\circ = 132.8^\circ.$$

Since $\angle ABC$ is obtuse,

$$\boxed{\angle ABC \approx 133^\circ \text{ (nearest degree).}}$$

Q20 (Exponential decay and rates)

$$P = 5000 b^{-t/10}, \quad b > 0, \quad P(20) = 1250.$$

Step 1: Find b

$$1250 = 5000 b^{-20/10} = 5000 b^{-2} \Rightarrow \frac{1}{4} = b^{-2} \Rightarrow b^2 = 4 \Rightarrow b = 2 \quad (b > 0).$$

So

$$P = 5000 \cdot 2^{-t/10}.$$

Step 2: Use the instantaneous rate of decrease

Write $P = 5000e^{(-t/10) \ln 2}$. Then

$$\frac{dP}{dt} = 5000e^{(-t/10) \ln 2} \left(-\frac{\ln 2}{10} \right) = -\frac{\ln 2}{10} P.$$

"Rate of decrease is 30" means $\frac{dP}{dt} = -30$:

$$-\frac{\ln 2}{10} P = -30 \Rightarrow P = \frac{300}{\ln 2}.$$

Now solve for t :

$$5000 \cdot 2^{-t/10} = \frac{300}{\ln 2} \Rightarrow 2^{-t/10} = \frac{3}{50 \ln 2}.$$

Taking $\log_2$:

$$-\frac{t}{10} = \log_2 \left(\frac{3}{50 \ln 2} \right) \Rightarrow t = 10 \log_2 \left(\frac{50 \ln 2}{3} \right).$$

$$t \approx 35.3.$$

$t \approx 35.3 \text{ years (to 1 d.p.)}$

Q21 (Rates and integration)

Initially $V(0) = 350$ L and

$$\frac{dV}{dt} = 300 - 7.5t, \quad t \text{ in hours.}$$

First find when $\frac{dV}{dt} = 0$:

$$300 - 7.5t = 0 \Rightarrow t = \frac{300}{7.5} = 40 \text{ h.}$$

Now find $V(t)$:

$$V = \int (300 - 7.5t) dt = 300t - 3.75t^2 + C.$$

Use $V(0) = 350 \Rightarrow C = 350$. Hence

$$V(t) = 300t - 3.75t^2 + 350.$$

So

$$V(40) = 300(40) - 3.75(40)^2 + 350 = 12000 - 3.75(1600) + 350 = 12000 - 6000$$

$V = 6350 \text{ L when } \frac{dV}{dt} = 0.$

Q22 (Geometric series – limiting sum)

Series: $2000 - 1200 + 720 - 432 + \dots$

$$a = 2000, \quad r = \frac{-1200}{2000} = -0.6.$$

Since $|r| < 1$, limiting sum exists:

$$S_{\infty} = \frac{a}{1-r} = \frac{2000}{1-(-0.6)} = \frac{2000}{1.6} = 1250.$$

$$S_{\infty} = 1250$$

Q23 (Annuities using the table)

(a) Save \$450 000 over 10 years at 6% p.a. compounded annually

From the table, FV factor for $n = 10$ at 6% is 13.181.

If the annual contribution is R (end of each year):

$$450000 = R(13.181) \Rightarrow R = \frac{450000}{13.181} \approx 34144.5.$$

$$R \approx \text{\$34,140 per year (nearest dollar)}$$

(b) Contribute \$8535 every 3 months for 10 years at 6% p.a. compounded quarterly

Quarterly rate $i = \frac{0.06}{4} = 0.015 = 1.5\%$. Number of quarters $n = 40$.

From the table, FV factor for $n = 40$ at 1.5% is 54.268.

$$FV = 8535(54.268) \approx 463177.38.$$

$$FV \approx \text{\$463 177.38 nearest dollar)}$$

Q24 – Teacher Solution (HSC style)

Let the curve be

$$y = ax^2 + bx + c, \quad a \neq 0.$$

At a particular point, the tangent and normal are:

$$t(x) = 2x + 3, \quad n(x) = -\frac{1}{2}x - 2.$$

Since the tangent and normal are at the **same point**, that point is the **intersection** of these two lines.

1) Find the point of tangency

$$2x + 3 = -\frac{1}{2}x - 2$$

$$2x + \frac{1}{2}x = -5 \Rightarrow \frac{5}{2}x = -5 \Rightarrow x = -2$$

Substitute into $t(x)$:

$$y = 2(-2) + 3 = -1$$

So the point is $\boxed{(-2, -1)}$.

2) Use the derivative (tangent gradient)

For $y = ax^2 + bx + c$,

$$\frac{dy}{dx} = 2ax + b.$$

At $x = -2$, the tangent gradient is 2, hence

$$2a(-2) + b = 2 \Rightarrow -4a + b = 2 \quad (1)$$

3) Use the turning point information

A quadratic has its turning point at

$$x = -\frac{b}{2a}.$$

Given the minimum turning point occurs at $x = -4$:

$$-\frac{b}{2a} = -4 \Rightarrow b = 8a \quad (2)$$

Substitute (2) into (1):

$$-4a + 8a = 2 \Rightarrow 4a = 2 \Rightarrow a = \frac{1}{2}.$$

Then

$$b = 8a = 8 \cdot \frac{1}{2} = 4.$$

4) Find c using the point $(-2, -1)$

$$-1 = a(-2)^2 + b(-2) + c = 4a - 2b + c$$

$$-1 = 4\left(\frac{1}{2}\right) - 2(4) + c = 2 - 8 + c \Rightarrow c = 5.$$

$$\boxed{a = \frac{1}{2}, \quad b = 4, \quad c = 5.}$$